FT8 Cluster Aggregator

for macOS

User Manual

Version 1.2.0

Aggregate FT8/FT4 spots from multiple WSJT-X/JTDX instances
and DX Cluster nodes into a unified telnet cluster server

Original Windows Application

Vinod VU3ESV / LB9KJ

macOS Version Conceptualised by

Manoj VU2CPL

macOS Version Developed with

Claude Code (Anthropic)

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1. Introduction

FT8ClusterAggregator is a macOS application designed for amateur radio operators. It collects FT8, FT4, and other digital mode spots from multiple sources and aggregates them into a unified DX cluster feed that can be consumed by logging software such as Logger32, DXLab, N1MM+, Log4OM, and others.

The application supports two types of spot sources:

- **UDP Sources** - Direct connections to WSJT-X, JTDX, or other software that broadcasts decoded FT8/FT4 spots via the WSJT-X UDP protocol
- **DX Cluster Nodes** - Telnet connections to traditional DX cluster servers (DX Spider, AR-Cluster, CC Cluster, etc.) to receive spots from the worldwide cluster network

All spots from all sources are combined and rebroadcast through a built-in telnet cluster server and/or UDP, allowing your logging software to receive a single aggregated stream of spots from multiple radios, SDRs, and cluster nodes.

This macOS version is a native Swift/SwiftUI port of the original Windows application created by Vinod VU3ESV/LB9KJ. The macOS version was conceptualised by Manoj VU2CPL.

2. System Requirements

Requirement	Details
Operating System	macOS 14.0 (Sonoma) or later
Architecture	Apple Silicon (M1/M2/M3/M4) and Intel
Disk Space	Less than 5 MB
Network	Local network access for UDP/TCP
Software	WSJT-X, JTDX, or compatible decoder (optional)

3. Installation

3.1 From Pre-built App Bundle

- Locate **FT8ClusterAggregator.app** in the distribution folder
- Drag it to your **/Applications** folder (or run from any location)
- On first launch, macOS may show a security warning since the app is not notarised
- Go to **System Settings > Privacy & Security** and click **"Open Anyway"**

Alternatively, open Terminal and run the following command to bypass Gatekeeper:

```
xattr -cr /path/to/FT8ClusterAggregator.app
```

Then right-click the app and select "Open" for the first launch. After that it will open normally with a double-click.

3.2 Building from Source

If you have the source code and Xcode Command Line Tools installed (install with: xcode-select --install):

Step 1: Clone and build

```
git clone https://github.com/vu2cpl/FT8ClusterAggregator-macOS.git
cd FT8ClusterAggregator-macOS
swift build -c release
```

3.3 Creating the .app Bundle

After building from source, you can create a proper macOS .app bundle that can be launched from Finder, placed in the Dock, or copied to /Applications:

Step 1: Create the bundle directory structure

```
mkdir -p FT8ClusterAggregator.app/Contents/MacOS
mkdir -p FT8ClusterAggregator.app/Contents/Resources
```

Step 2: Copy the binary and icon

```
cp .build/release/FT8ClusterAggregator FT8ClusterAggregator.app/Contents/MacOS/
cp AppIcon.icns FT8ClusterAggregator.app/Contents/Resources/
echo -n "APPL????" > FT8ClusterAggregator.app/Contents/PkgInfo
```

Step 3: Create the Info.plist

Create the file FT8ClusterAggregator.app/Contents/Info.plist with the following content (a template is provided in the project README):

Key	Value
CFBundleName	FT8ClusterAggregator
CFBundleDisplayName	FT8 Cluster Aggregator
CFBundleIdentifier	com.vu2cpl.ft8clusteraggregator
CFBundleVersion	1.2.0
CFBundleExecutable	FT8ClusterAggregator
CFBundlePackageType	APPL
CFBundleIconFile	AppIcon
LSMinimumSystemVersion	14.0
NSHighResolutionCapable	true

The full Info.plist XML template is available in the project README on GitHub.

Step 4: Sign and launch

```
codesign --force --deep --sign - FT8ClusterAggregator.app
open FT8ClusterAggregator.app
```

Step 5 (Optional): Install to Applications

```
cp -r FT8ClusterAggregator.app /Applications/
```

If sharing the built .app with others, they will need to run xattr -cr /path/to/FT8ClusterAggregator.app and right-click > Open on the first launch, as the app is not notarised through the Apple Developer Program.

4. Getting Started

4.1 Main Window Overview

When you launch FT8ClusterAggregator, you will see the main window with the following sections from top to bottom:

Section	Purpose
Header	App name and version number
Configuration	Callsign, TCP cluster port, and broadcast destinations
UDP Sources	List of WSJT-X/JTDX UDP sources with IP, port, and status
DX Cluster Nodes	List of telnet DX cluster sources with address, credentials, and status
Controls	CQ-Only filter, Minimize on Start, Clear Spots, Start/Stop Monitoring
Spots Table	Real-time display of aggregated spots from all sources
Status Bar	Connection status, active source count, and total spot count

4.2 Setting Your Callsign

Enter your callsign in the **Callsign** field at the top of the window. This callsign is used as the spotter identifier when rebroadcasting spots through the cluster server. The callsign is automatically converted to uppercase. It is also used as the default username when adding new DX Cluster sources.

5. Configuring UDP Sources (WSJT-X / JTDX)

UDP Sources receive decoded FT8/FT4/JT65/JT9 spots directly from WSJT-X, JTDX, or any software that implements the WSJT-X UDP protocol. The app parses the binary WSJT-X protocol messages (Status and Decode types) to extract spot information.

5.1 Adding a UDP Source

Click **"Add UDP Source"** in the UDP Sources section. Configure the following fields:

Field	Description	Default
Name	A label to identify this source (e.g., "WSJT-X 20m")	Source N
Listen IP	IP to listen on. Use 0.0.0.0 for all interfaces	0.0.0.0
Port	UDP port number matching your WSJT-X/JTDX setting	2237
Enabled	Toggle to enable/disable this source	On

Note: In WSJT-X, go to File > Settings > Reporting and set the UDP Server address to 127.0.0.1 (or your Mac's IP if running on another machine) and the port to match the port configured here.

5.2 Multiple Instances

You can add multiple UDP sources to aggregate spots from several WSJT-X/JTDX instances running simultaneously. Each instance must be configured to broadcast on a different UDP port. For example:

- WSJT-X on 20m band: UDP port 2237
- JTDX on 40m band: UDP port 2238
- WSJT-X on 15m band: UDP port 2239

6. Configuring DX Cluster Sources

DX Cluster sources connect to traditional DX cluster telnet servers to receive spots from the worldwide amateur radio spotting network. These spots are aggregated alongside your local WSJT-X/JTDX decodes.

6.1 Adding a DX Cluster Node

Click **"Add DX Cluster"** in the DX Cluster Nodes section. Configure the following fields:

Field	Description	Default
Name	A label for this cluster (e.g., "VE7CC")	Cluster N
Address	Hostname or IP of the cluster node	(empty)
Port	Telnet port number	7300
Username	Your callsign for login	Your callsign
Password	Password if required (many clusters do not require one)	(empty)
Enabled	Toggle to enable/disable	On

Popular DX Cluster Nodes

Node	Address	Port
VE7CC	dxc.ve7cc.net	23
W9PA	dxc.w9pa.net	7300
GB7DXC	gb7dxc.dxcluster.co.uk	7300
WA9PIE	dxc.wa9pie.net	7300
K1TTT	k1ttt.net	7300

6.2 Authentication

The application automatically detects login and password prompts from the cluster server. Most DX cluster nodes only require a callsign (no password). The app sends your username when it detects a login prompt, and your password (if provided) when it detects a password prompt. Connection status is shown next to each cluster entry:

- **Connecting...** - TCP connection is being established
- **Connected** - TCP connection established, waiting for login
- **Authenticated** - Successfully logged in and receiving spots

7. Broadcast Destinations

Aggregated spots are rebroadcast through two mechanisms:

7.1 UDP Broadcast

Two UDP broadcast destinations can be configured. Each has an IP address and port. Spots are sent as DX cluster-formatted text lines via UDP. Click **Save** after changing the settings. These can be used to feed spots to other applications on the local machine or across the network.

7.2 TCP Telnet Cluster Server

The app runs a built-in TCP telnet cluster server. Any logging software that supports connecting to a DX cluster via telnet can connect to this server to receive the aggregated spot feed. The default port is **7550**.

Configure your logging software to connect to:

- **Host:** 127.0.0.1 (if on the same machine) or your Mac's IP address
- **Port:** 7550 (or whatever you configured)

The status bar shows the number of connected telnet clients.

8. Monitoring & Spot Aggregation

8.1 Starting Monitoring

Click the green **"Start Monitoring"** button to begin. This will:

- Start a UDP listener for each enabled UDP source
- Connect to each enabled DX Cluster node
- Start the TCP telnet cluster server
- Configure the UDP broadcast destinations

*While monitoring is active, source settings cannot be modified. Click the red **"Stop Monitoring"** button to stop and reconfigure.*

8.2 CQ-Only Filter

Enable the **CQ Only** toggle to filter spots so that only CQ calls are displayed and rebroadcast. This reduces the volume of spots to only stations calling CQ, which are the ones available to work. QSO exchanges (reports, RRR, 73) are filtered out.

8.3 Understanding the Spots Table

The spots table displays all aggregated spots in real-time:

Column	Description
Time	UTC time of the spot (HHmm format)
Source	Name of the source that provided this spot
Callsign	The DX station callsign extracted from the message
Freq (MHz)	Frequency in MHz (dial frequency + audio offset)
SNR	Signal-to-noise ratio in dB (from WSJT-X decodes)
Mode	Operating mode (FT8, FT4, CW, SSB, etc.)
Message	Full decoded message or spot comment

CQ spots are highlighted with a green background in the table for easy identification.

9. Connecting Logging Software

FT8ClusterAggregator acts as a telnet DX cluster server. To connect your logging software:

Logger32

- Go to Setup > Telnet > DX Cluster
- Set Host to **127.0.0.1** and Port to **7550**

- Click Connect

N1MM+

- Go to Window > Telnet
- Enter **127.0.0.1:7550** as the cluster address

Log4OM

- Go to Settings > Cluster
- Add a new cluster with address **127.0.0.1** port **7550**

Any Telnet Client

You can also test the cluster server using a terminal:

```
telnet 127.0.0.1 7550
```

10. Troubleshooting

No spots appearing from WSJT-X

Verify the UDP port in WSJT-X Settings > Reporting matches the port configured in FT8ClusterAggregator. Ensure WSJT-X is set to broadcast to 127.0.0.1 (or your Mac's IP). Check that the source is enabled and shows "Active" status.

DX Cluster not connecting

Verify the cluster address and port are correct. Check your internet connection. Some clusters may be down - try a different node. Check if a firewall is blocking outbound TCP connections.

DX Cluster shows "Waiting..."

The cluster node may be unreachable or the DNS lookup failed. Try using an IP address instead of a hostname.

macOS security warning on launch

Go to System Settings > Privacy & Security and click "Open Anyway". This is needed because the app is not notarised through the Apple Developer Program.

Logging software cannot connect

Ensure monitoring is started (green "Start Monitoring" button). Verify the TCP cluster port (default 7550) is not blocked by a firewall. Check that the port is not already in use by another application.

Port already in use

Another application is using the same port. Either close that application or change the port in FT8ClusterAggregator to an unused port number.

Spots from cluster but no frequency

DX Cluster spots include frequency information from the cluster server. This is normal - the frequency comes from the spotter, not from your radio.

11. Credits & Acknowledgements

Original Windows Application

The original FT8ClusterAggregator was developed as a Windows .NET application by **Vinod, VU3ESV / LB9KJ**. His vision of aggregating FT8 spots from multiple sources into a unified DX cluster feed laid the foundation for this project.

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macOS Version

The macOS version of FT8ClusterAggregator was conceptualised by **Manoj, VU2CPL**. This native Swift/SwiftUI port brings the functionality of the original Windows application to the Mac platform, with enhancements including multiple UDP source support, DX Cluster telnet node integration, and a modern macOS user interface.

Development

The macOS application was developed with the assistance of **Claude Code** by Anthropic, an AI-powered software engineering tool.

Open Source Libraries & Protocols

- **WSJT-X UDP Protocol** - Joe Taylor K1JT and the WSJT-X development team
- **DX Cluster Protocol** - DX Spider and AR-Cluster communities
- **Apple Network Framework** - For TCP/UDP networking on macOS
- **SwiftUI** - Apple's modern UI framework for macOS

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